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AI LAB 1.py - C:/Users/K Aishwarya/Documents/AI LAB 1.py (3.14.2)
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from collections import deque

def solve_puzzle(start, goal):
    queue = deque([start, []])
    visited = [start]

    moves = [(-1, 0), (1, 0), (0, -1), (0, 1)]

    while queue:
        state, path = queue.popleft()
        if state == goal:
            return path + [state]

        zero = state.index(0)
        row = zero // 3
        col = zero % 3

        for dr, dc in moves:
            new_row = row + dr
            new_col = col + dc

            if 0 <= new_row < 3 and 0 <= new_col < 3:
                new_zero = new_row * 3 + new_col

                new_state = list(state)
                new_state[zero], new_state[new_zero] = \
                    new_state[new_zero], new_state[zero]

                new_state = tuple(new_state)

                if new_state not in visited:
                    visited.add(new_state)
                    queue.append((new_state, path + [state]))

    return None

start = (1, 2, 3,
        4, 0, 6,
        7, 5, 8)

goal = (1, 2, 3,
        4, 5, 6,
        7, 8, 0)

solution = solve_puzzle(start, goal)
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>>> ===== RESTART: C:/Users/K Aishwarya/Documents/AI LAB 1.py =====
Solution:
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```